PartSight: Identifying Marine Diesel Engine Parts from Photos with AlexNet

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Problem Statement

Ship engineers often send suppliers a photo of a worn generator-engine part with no label or part number. Identifying it by eye is slow, depends on a few experts, and errors lead to wrong orders and longer vessel downtime.

Goal: given one photo, predict which of six high-volume parts it shows.

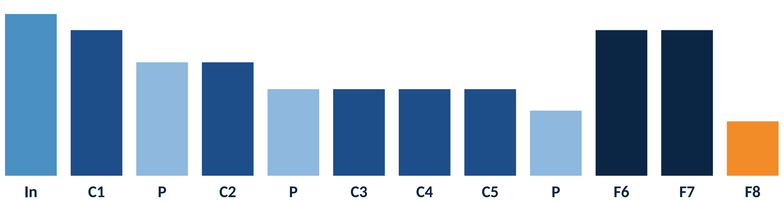
- **Fuel injector**
- **Turbocharger**
- **Piston**
- **Connecting rod**
- **Crankshaft**
- **Oil filter**

Methods

Dataset

- 60 web image searches (10 per part) → 1,695 photos downloaded
- CLIP filter removed off-topic photos and diagrams; near-duplicates removed
- 1,101 clean photos, split 70% train / 15% val / 15% test

AlexNet (Krizhevsky et al., 2012)



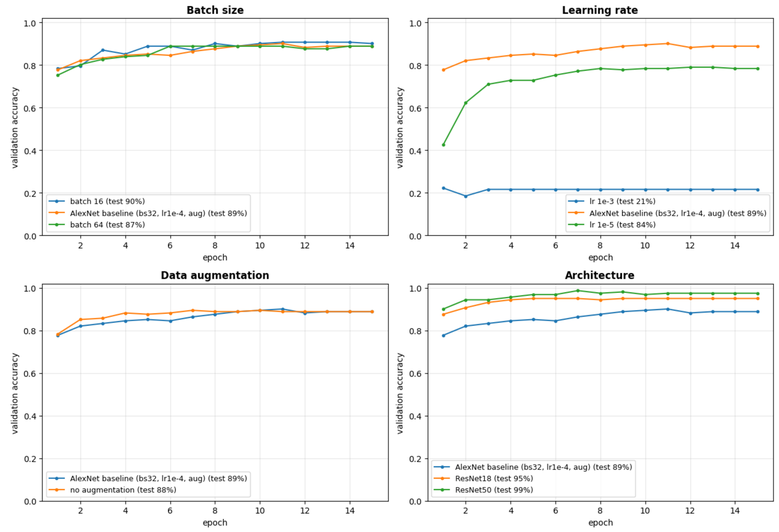
5 conv layers (C) with max-pooling (P), 3 fully connected layers (F). ImageNet-pretrained; last layer → 6 parts.

- 224×224 input; random crop, flip, rotation, color jitter
- Cross-entropy (label smoothing 0.1), Adam, cosine LR, 15 epochs
- Best epoch chosen on validation; scored once on test
- PyTorch on Colab T4 GPU; runs logged to Weights & Biases

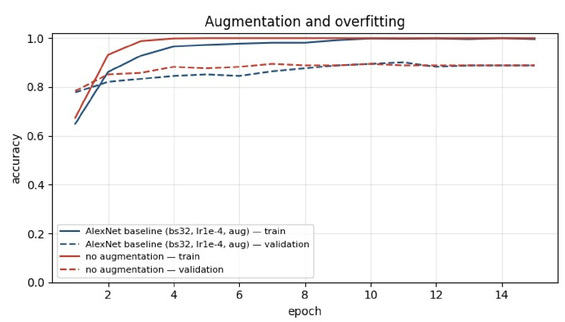
Experiments & Results

Each run changes one setting from the AlexNet baseline (batch 32, LR 1e-4, augmentation on).

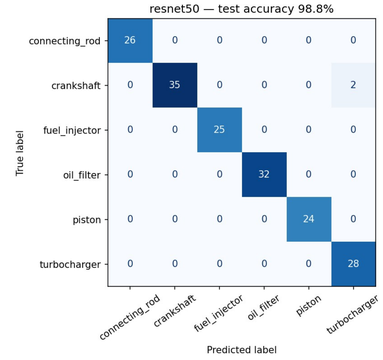
Run	Change	Test acc.
baseline	—	89.0%
bs16	batch 16	**90.1%**
bs64	batch 64	87.2%
lr1e-3	LR 1e-3	20.9%
lr1e-5	LR 1e-5	84.3%
no_aug	no augmentation	88.4%
resnet18	ResNet18	95.3%
resnet50	ResNet50	**98.8%**



Validation accuracy per epoch for each experiment group (test accuracy in the legend).



Without augmentation (red) training accuracy hits 100% by epoch 4: the model memorizes.



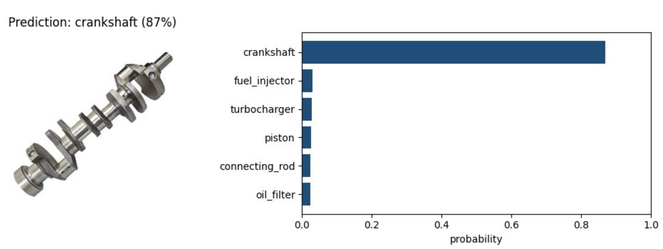
Best model (ResNet50) on 172 held-out test photos: 2 errors.

Discussion

- Learning rate mattered most: 1e-3 wiped out the pretrained weights (20.9%, about guessing); 1e-5 learned too slowly (84.3%).
- Batch size had a small effect: batch 16 was the best AlexNet (90.1%).
- Augmentation slowed memorization; the test gain was small (89.0% vs 88.4%).
- ResNet50 (98.8%) and ResNet18 (95.3%) beat AlexNet in the same ~4 minutes.
- Data quality was the biggest lever: CLIP cleaning + more photos lifted ResNet50 from 89.6% to 98.8%.

Conclusions & Future Work

- Best AlexNet 90.1%; best overall ResNet50 98.8%
- Clean data and one-change experiments mattered more than any single setting
- Next: real engine-room photos, more parts, part-number prediction, phone app



A new photo the model never saw: crankshaft, 87% confidence.

[1] Krizhevsky, Sutskever & Hinton (2012). ImageNet Classification with Deep CNNs. NeurIPS. [2] He et al. (2016). Deep Residual Learning. CVPR. [3] Radford et al. (2021). CLIP. ICML.